deep sclerectomy. In two eyes, percolation through the trabeculodescemetic membrane was weak, so the procedure was supplemented by a trabeculotomy. In another four eyes, identification of Schlemm's canal was not made, so no deep sclerectomy alone nor added trabeculotomy could be done, and the procedure was converted into a guarded filtering procedure. Finally, in two eyes an inadvertent visible perforation of the trabeculodescemetic membrane occurred. The report of a high percentage of conversion to perforating filtering surgery and potential associated postoperative complications differs from our own experience in refractory congenital glaucoma. After both pure deep sclerectomies, hyphemas were reported as complications, although trivial. These may further result from possible microperforations, with a resultant high and abrupt change in IOP. We agree that deep sclerectomy alone is not applicable in most cases with complicated refractory congenital glaucomas. We make small visible perforations of the trabeculodescemetic membrane with the tip of a 30-gauge needle. Our incidence of nonidentified Sclemm's canal in congenital refractory glaucomas, however, is low.

Considering that only pure nonpenetrating filtering surgery is reported in this article, the IOP control after 19 months of follow-up was 50%, which, considering the high-risk failure rate for refractory congenital glaucoma, is not so bad.

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Author reply

Dear Editor:
Drs. Ravinet and Mermoud detailed their deep sclerectomy techniques for treatment of refractory congenital glaucomas in their comment on our retrospective case series.

Between our observation and the experience of Drs. Ravinet and Mermoud there is complete consensus that nonpenetrating deep sclerectomy alone is not applicable in most cases with refractory congenital glaucoma.

To enhance aqueous outflow after deep sclerectomy they needle the trabeculodescemetic membrane, which causes macroperforations. Additionally, to allow subconjunctival filtration, only loose readaptation of the superficial scleral flap is performed. In our opinion this technique is a variant of filtering surgery or, as we would term it, a "penetrating deep sclerectomy."

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Sildenafil-Associated NAION

Dear Editor:
Pomeranz and colleagues' case series describing the occurrence of nonarteritic anterior ischemic optic neuropathy (NAION) after ingestion of sildenafil in five subjects invites speculation about the causative mechanism.[1] Their thesis primarily emphasized the notions of disrupted autoregulation in the optic nerve head (ONH) or a toxic optic neuropathy from potentiation of nitric oxide by sildenafil. Importantly, although their five patients had crowded discs, mention of systemic arterial hypotension as a possibile etiology was omitted. Systemic hypotension is particularly relevant in the context of a structurally congested ONH that has vulnerable watershed zones in its microcirculation.[2]

In this regard, the manufacturer (Pfizer Pharmaceuticals, New York, NY) has reported that sildenafil reduces blood pressure (independent of the amount of drug) in healthy supine subjects, especially 1 to 2 hours after administration. Subsequent studies have confirmed that sildenafil can produce a small decrease in systolic and diastolic blood pressure, with some groups finding a fall as large as 20 mmHg (systolic) in certain individuals, suggesting perhaps a difference in interindividual sensitivity to the drug.[3,4] Some workers examining the hemodynamic effects of sildenafil have used conventional sphygmomanometry at the brachial artery to assess effects on blood pressure. It is rather thought provoking that sphygmomanometry may significantly underestimate hemodynamic disturbances (e.g., the fall in aortic systolic and pulse pressures) induced by vasodilator agents when compared with relatively sophisticated techniques.[5] Therefore, hypotension and other hemodynamic changes in the central circulation that affect ONH perfusion after vasodilator therapy may be overlooked when attention is focused on detecting changes in brachial arterial pressure with sphygmomanometry.[6] These transient fluxes in the central circulation (though not florid enough to produce acute symptomatic hypotension) may be sufficient to elicit the final insult of critical ischemia in the anatomically susceptible ONH.

The role of arterial hypotension deserves explicit consideration in these five patients, where crowded ONHs were exposed to a drug known to modify systemic hemodynamics. Moreover, an allusion to this established mechanism does not invalidate Pomeranz et al's proposition that sildenafil might interfere with regulation of ONH microvasculature, because both local and systemic circulatory effects could conspire in the pathogenesis. Their alternative theory of a toxic optic neuropathy from sildenafil is less compelling because all five patients have clinical and investigative evidence compatible with classic NAION.

It is difficult to be specific about the mechanisms responsible for NAION secondary to sildenafil because rigid statements cannot be supported by existing evidence. Certainly, we cannot be dogmatic or claim omniscience because the pathophysiology of NAION is still poorly characterized. Suffice it to say, there are many predisposing and precipitating factors for NAION, and various permutations of these risks can generate ischemia at the ONH.[7] Sildenafil (like other antihypertensives or, indeed, like other acquired risks) may serve as a trigger given a strong enough predisposition.

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